UNLOCKING THE *Archive*

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Context

A major European digitisation programme recently preserved thousands of culturally significant videotapes at risk of loss. Many files arrived without reliable metadata. Archivists cannot tell, by inspection, whether a file contains one broadcast or several, nor what topics it covers.

Problem

Manual cataloguing at this scale is prohibitively time-consuming, leaving vast portions of audiovisual heritage effectively inaccessible.

"Computational moving image studies must work with the grain of the archive — not against its cataloguing realities."

SITUATED WITHIN ARNOLD & TILTON 2023; CHÁVEZ HERAS 2024; RETTBERG 2023.

Approach

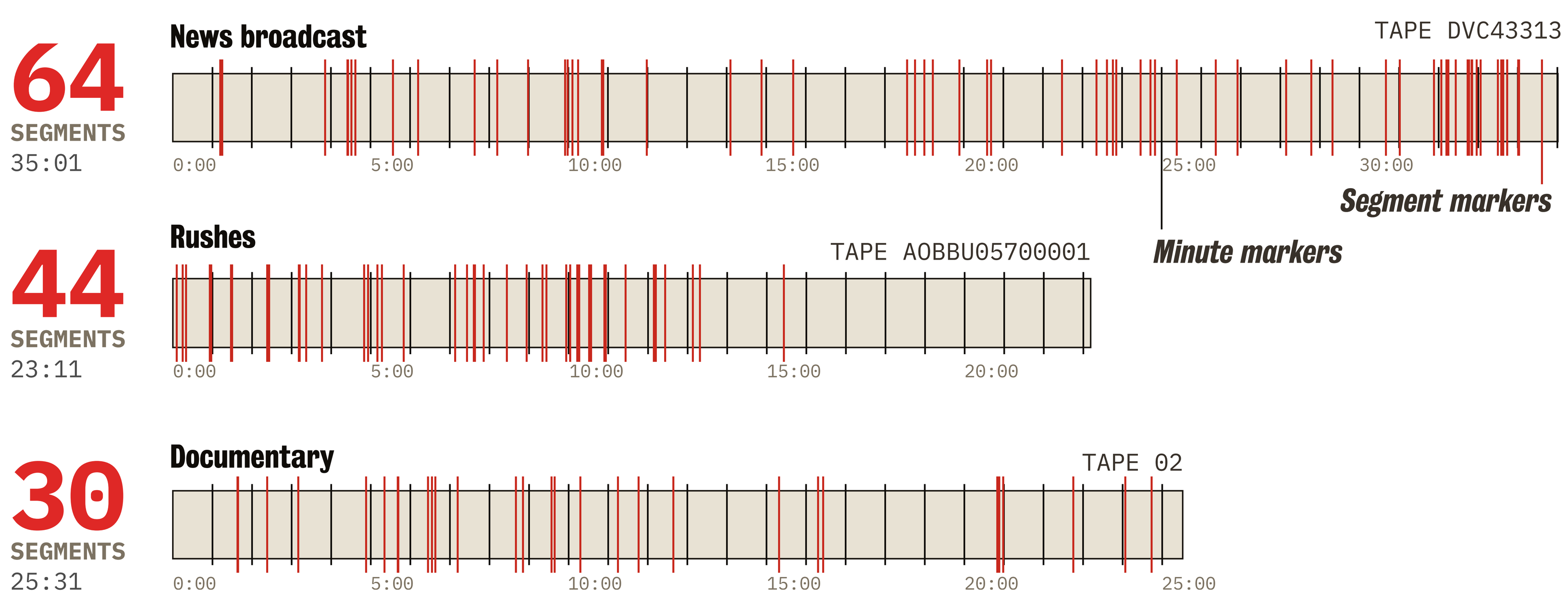
We present a tool that automatically identifies content boundaries, classifies segment types, extracts topics, and generates summaries. Unlike shot-boundary detectors that find technical cuts, our approach targets semantic boundaries, where one topic ends and another begins.

The pipeline is local-first: open-weight models, consumer hardware, air-gap friendly. The default configuration ensures sensitive material never leaves the institution's infrastructure.

A partner institution repurposed the same pipeline for advertisement detection across 50 videos — a different task altogether.

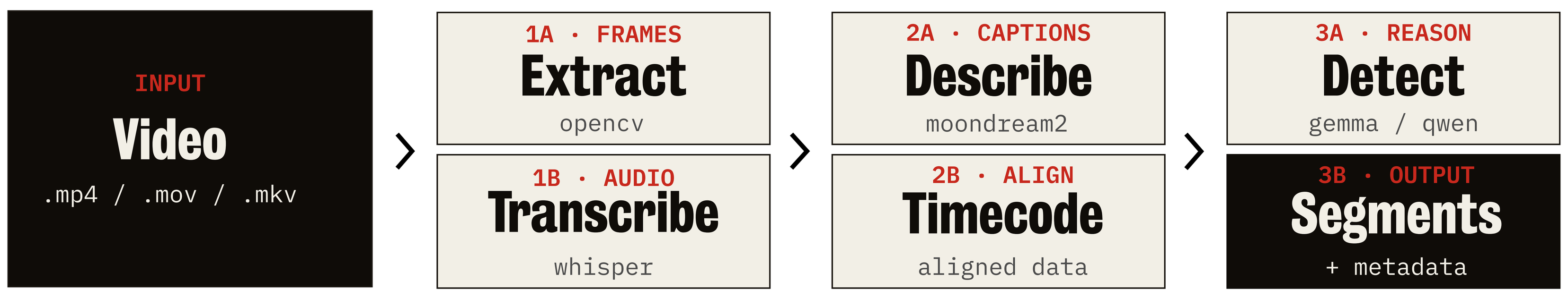
Three corpora, one pipeline

A "distant view" of every boundary the model detected across our three test tapes. Each red tick marks a semantic segment boundary; black ticks mark the minute axis.



How we see without watching

Three modalities aligned by timestamp, then reasoned over jointly. The architecture is model-agnostic, a partner institution ran the same code path with larger local models.



1 Visual

MOONDREAM2 · 1.9B PARAMS · 8GB VRAM

Generates natural-language descriptions of sampled frames — capturing studio settings, on-screen text, and presenter changes.

2 Audio

OPENAI WHISPER · TIMESTAMPED

Extracts timestamped speech, providing the narrative thread that often defines segment boundaries.

3 Reasoning

GEMMA-3-4B DEFAULT · QWEN TESTED

A local LLM reads the aligned captions and transcript in a 3-frame sliding window (previous / current / next) to detect semantic boundaries.

What the pipeline saw

Two of the three test tapes, presented as contact sheets. Each thumbnail is a sampled frame; the yellow chip is the segment number the boundary detector assigned.

Read left-to-right, top-to-bottom.

Documentary · single-presenter, location work, 25 min reel. 30 Segments.

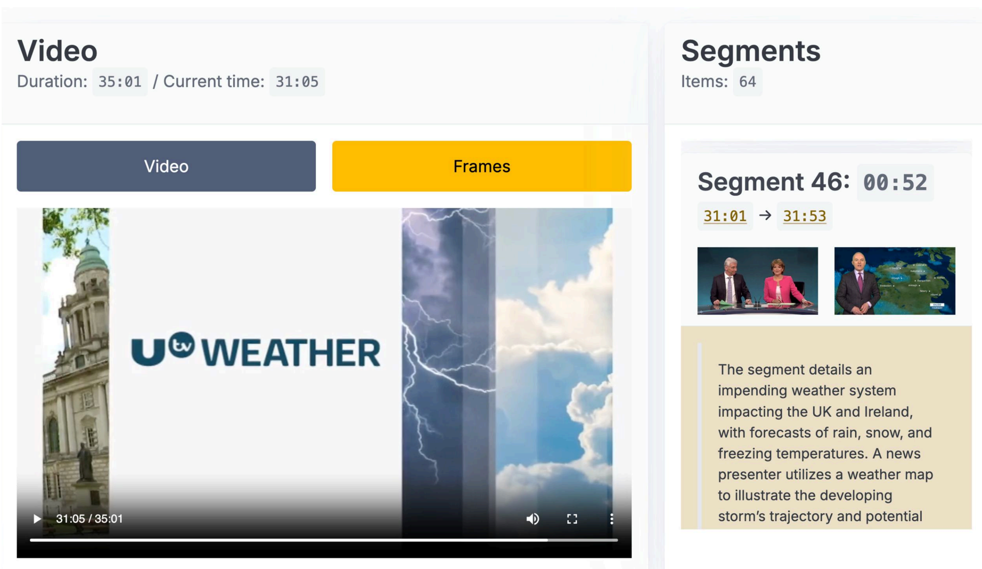
Rushes · off-air broadcast, unedited, 23 min reel. 44 Segments.

Verification

An offline dashboard displays the video alongside segment cards for human verification. No server. Air-gap friendly. The active segment highlights as the video plays; clicking any timestamp jumps the head. Two display modes, full video, or the contact-sheet view of every sampled frame.

Every output file carries the same provenance fingerprint:

MODEL NAME | PROCESSING TIME | DEVICE | GIT HASH | SOFTWARE VERSIONS | PLATFORM | HOSTNAME



The verification dashboard at 31:05 / 35:01 of the news broadcast. Segment 46 is auto-selected; its two frames and free-text summary are shown on the right.

Findings

87 min / 35 min tape On a consumer GPU. Frame captioning dominates the runtime; the rest is I/O. Bounded by silicon, not by budget.

Reproducible Each output JSON ships with model name, git hash, device, processing time, software versions, platform — a clean audit trail.

Known limitation Short spurious segments (1–5 s) occur when the model reacts to brief visual changes. The 3-frame window mitigates this. From an archival standpoint this over-segmentation is largely acceptable, finer segments improve discoverability and access, which is the primary goal. Formal ground-truth evaluation is in progress.

Resources

Further information about the project, and underlying research is available

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